

```
In [49]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
```

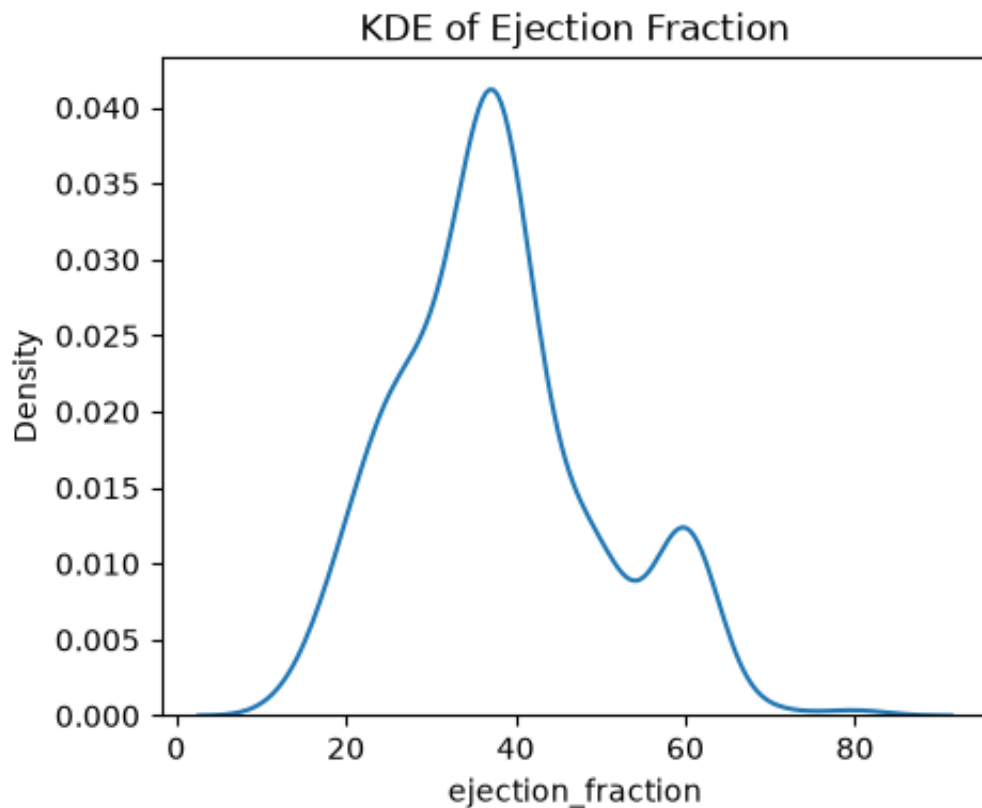
Problem 4a

```
In [54]: df = pd.read_csv('data/heart_failure_clinical_records_dataset.csv')
df.head()
```

```
Out[54]:
```

	age	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_k
0	75.0	0	582	0	20	
1	55.0	0	7861	0	38	
2	65.0	0	146	0	20	
3	50.0	1	111	0	20	
4	65.0	1	160	1	20	

```
In [71]: plt.figure(figsize=(5, 4))
sns.kdeplot(df['ejection_fraction'])
plt.title('KDE of Ejection Fraction')
plt.show()
```



Problem 4b

```
In [72]: plt.figure(figsize=(5, 4))
         for i in range(15):
             df_sample = df.sample(frac=1.0, replace=True)
             sns.kdeplot(df_sample['ejection_fraction'], color='gray')

         # Original plot overlayed for reference
         sns.kdeplot(df['ejection_fraction'], color='black')
```

Out[72]: <Axes: xlabel='ejection_fraction', ylabel='Density'>

